# Supplemental Material for Dimension-dependent minimax sensitivity for measuring quantum imaginarity

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3 November 2026

## 1 Scope and reproducibility

This supplement provides the numerical details for the finite-dimensional examples and additional structural results for five-outcome qutrit POVMs. The exact minimax theorem, the finite-outcome threshold, the approximation bounds, and the global four-outcome qutrit optimum are proved in the main article. The additional structural results below characterize rank-one lifts, zero-defect designs, and parts of the rank-two stratum. They do not establish the global five-outcome optimum; the five-outcome POVM in the main text is an explicit feasible improvement over the four-outcome optimum, not a claimed global optimizer.

## 2 Exact five-outcome qutrit construction

For completeness, the isometry used in the main text is

$$Q_0 = \begin{pmatrix} \frac{1}{2\sqrt{2}} & -\frac{1+i\sqrt{3}}{4\sqrt{2}} & -\frac{1-i\sqrt{3}}{4\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & -\frac{1+i\sqrt{3}}{4\sqrt{2}} & -\frac{1-i\sqrt{3}}{4\sqrt{2}} \\ \frac{1}{2} & \frac{1+i\sqrt{3}}{4} & \frac{1-i\sqrt{3}}{4} \\ \frac{1}{2} & \frac{1+i\sqrt{3}}{4} & \frac{-1+i\sqrt{3}}{4} \\ -\frac{1}{2} & \frac{1+i\sqrt{3}}{4} & \frac{-1+i\sqrt{3}}{4} \end{pmatrix}.$$

For an ordered pair of distinct indices $i, j \in \{1, \ldots, 5\}$, let $G_{ij}(\theta, \phi)$ be the identity except for the block on the ordered coordinates $(i, j)$, where it equals

$$\begin{pmatrix} \cos\theta & e^{i\phi}\sin\theta \\ -e^{-i\phi}\sin\theta & \cos\theta \end{pmatrix}.$$

Set

$$Q = G_{34}\Big(\frac{\pi}{41}, \frac{\pi}{2}\Big)\, G_{41}\Big(\frac{\pi}{66}, \pi\Big)\, G_{23}\Big(-\frac{\pi}{44}, 0\Big)\, G_{25}\Big(-\frac{\pi}{29}, \frac{3\pi}{2}\Big)\, Q_0.$$

Then $Q^\dagger Q = I_3$, and the effects $E_k = q_k^\dagger q_k$ formed from the rows $q_k$ of $Q$ give an exact rank-one five-outcome qutrit POVM. For the Cartesian spin-one generators $(J_a)_{mn} = -i\epsilon_{amn}$, put $\mathbf{b}_k = (\operatorname{Tr} E_k J_x, \operatorname{Tr} E_k J_y, \operatorname{Tr} E_k J_z)$. The ten facet sensitivities are listed in Table S1.

The minimum occurs uniquely at $(2, 3)$ and lies between 0.620509845202 and 0.620509845203. Every cross-product norm has a positive lower enclosure, and every facet is strictly above $\sqrt{3/8}$. The analytic isometry proof supplies exact normalization; numerical intervals are not used to infer an exact identity from a small residual. The full machine-readable record is `five_outcome_interval_certificate_v53.json`.

Table S1: Outward interval enclosures of the ten facet sensitivities. The interval calculation uses 80 decimal digits; the displayed endpoints are rounded outward to 12 decimal places and verified by interval comparisons.

| Pair $(i, j)$ | Certified open enclosure |
|---|---|
| $(2, 3)$ | $(0.620509845202,\ 0.620509845203)$ |
| $(4, 5)$ | $(0.620548480327,\ 0.620548480328)$ |
| $(1, 4)$ | $(0.620597054558,\ 0.620597054559)$ |
| $(2, 4)$ | $(0.620632175113,\ 0.620632175114)$ |
| $(3, 5)$ | $(0.621051773202,\ 0.621051773203)$ |
| $(1, 3)$ | $(0.622309271434,\ 0.622309271435)$ |
| $(2, 5)$ | $(0.627185082614,\ 0.627185082615)$ |
| $(1, 5)$ | $(0.628634882840,\ 0.628634882841)$ |
| $(3, 4)$ | $(0.629143086467,\ 0.629143086468)$ |
| $(1, 2)$ | $(0.692229025342,\ 0.692229025343)$ |

## 3 Polyhedral qutrit checks

The reproduction script evaluates the zonotope facet formula for the tetrahedral, icosahedral, and dodecahedral spin-one POVMs. The results are shown in Table S2.

Table S2: Numerical sensitivities of the polyhedral qutrit POVMs. The analytic values are given in the main text.

| POVM | Worst-case | Peak |
|---|---|---|
| Tetrahedral, 4 outcomes | 0.612372435695795 | 0.866025403784439 |
| Icosahedral, 12 outcomes | 0.688190960235587 | 0.809016994374948 |
| Dodecahedral, 20 outcomes | 0.716978135976772 | not used |

For the dodecahedral case, the 45 unoriented pair-normal classes split into two symmetry classes of 30 and 15 directions. Their support values are the two exact expressions given in the main text.

## 4 Continuous-frame numerical sanity check

For the canonical skew-symmetric matrix with equal nonzero singular values, the script samples 200 000 independent ordered orthonormal two-frames in each dimension $2 \le d \le 8$. The empirical ratios agree with the analytic constants within 1.7 estimated standard errors for $d \ge 3$. In $d = 2$ the integrand is deterministic. This test checks normalization and implementation only; the exact minimax formula is proved analytically in the main article.

## 5 Additional structure of five-outcome qutrit POVMs

The results in this section are supplementary structural statements. They provide parameterizations and stability bounds, but they do not identify the unrestricted global five-outcome optimum.

### 5.1 A finite facet formula for five generators

**Proposition 5.1** (Finite facet formula for a five-generator qutrit zonotope)**.** *Let* $\mathbf{b}_1, \dots, \mathbf{b}_5 \in \mathbb{R}^3$ *span* $\mathbb{R}^3$ *and satisfy* $\sum_{k=1}^{5} \mathbf{b}_k = 0$*. Define*

$$h(\mathbf{a}) = \frac{1}{2} \sum_{k=1}^{5} |\mathbf{a} \cdot \mathbf{b}_k|, \qquad \|\mathbf{a}\|_2 = 1.$$

*For every pair* $i < j$ *with* $\mathbf{b}_i \times \mathbf{b}_j \neq 0$*, set*

$$\mathbf{n}_{ij} = \frac{\mathbf{b}_i \times \mathbf{b}_j}{\|\mathbf{b}_i \times \mathbf{b}_j\|_2}.$$

*Then the inradius of the associated zonotope is*

$$\min_{\|\mathbf{a}\|_2=1} h(\mathbf{a}) = \min_{\substack{i<j \\ \mathbf{b}_i \times \mathbf{b}_j \neq 0}} \max_{k \notin \{i,j\}} \frac{|\det(\mathbf{b}_i, \mathbf{b}_j, \mathbf{b}_k)|}{\|\mathbf{b}_i \times \mathbf{b}_j\|_2}. \tag{1}$$

*Proof.* The zonotope $Z = \sum_{k=1}^{5} [-\mathbf{b}_k/2, \mathbf{b}_k/2]$ is a full-dimensional centrally symmetric polytope. Its inradius is the minimum of its support function over the unit sphere, and this minimum is attained at a facet normal. A facet normal is perpendicular to at least two nonparallel generators, so it is parallel to some $\mathbf{b}_i \times \mathbf{b}_j$. For this normal, put $x_k = \mathbf{n}_{ij} \cdot \mathbf{b}_k$. Since the generators are centered, $\sum_k x_k = 0$, while $x_i = x_j = 0$. Thus the remaining three real numbers sum to zero, which implies $\frac{1}{2} \sum_{k \notin \{i,j\}} |x_k| = \max_{k \notin \{i,j\}} |x_k|$. Finally, $|x_k| = |\det(\mathbf{b}_i, \mathbf{b}_j, \mathbf{b}_k)| / \|\mathbf{b}_i \times \mathbf{b}_j\|_2$. Taking the minimum over all facet normals proves the formula. Redundant pairs are omitted when generators are parallel.

### 5.2 Rank-one effect lifts and the anisotropy budget

**Proposition 5.2** (A positivity bound for the qutrit imaginarity vector)**.** *Let* $E \geq 0$ *be a qutrit effect, write* $E = A + iB$ *with* $A = A^T \in \mathbb{R}^{3\times3}$ *and* $B = -B^T \in \mathbb{R}^{3\times3}$*, and define* $b_a = \mathrm{Tr}(EJ_a)$ *for the spin-one generators* $(J_a)_{mn} = -i\epsilon_{amn}$*. Put* $t = \|\mathbf{b}\|_2$*. If* $t > 0$ *and* $\mathbf{n} = \mathbf{b}/t$*, then*

$$\mathrm{Tr}(E) - t \geq \mathbf{n}^T A \mathbf{n} \geq 0. \tag{2}$$

*Equivalently, the trace defect of an effect controls its real-symmetric weight along the direction of its imaginarity vector.*

*Proof.* The matrix $iB$ is Hermitian and has eigenvalues $-t/2, 0, t/2$. Its eigenvectors for the two nonzero eigenvalues are circular vectors in the plane orthogonal to $\mathbf{n}$. Choose a normalized eigenvector $|v_-\rangle$ with $iB|v_-\rangle = -(t/2)|v_-\rangle$. Its real and imaginary parts are orthogonal, have equal norm, and span the plane perpendicular to $\mathbf{n}$. Consequently,

$$\langle v_-|A|v_-\rangle = \frac{1}{2}\big[\mathrm{Tr}(A) - \mathbf{n}^T A \mathbf{n}\big].$$

Since $E = A + iB \geq 0$,

$$0 \leq \langle v_-|E|v_-\rangle = \frac{1}{2}\big[\mathrm{Tr}(A) - \mathbf{n}^T A \mathbf{n} - t\big],$$

which is Eq. (2). The second inequality follows from $A \geq 0$, because $A$ is the restriction of the positive effect $E$ to real vectors.

**Proposition 5.3** (A block-positivity refinement of the effect defect)**.** *Under the assumptions of Proposition 5.2, choose a real orthogonal basis whose first vector is* $\mathbf{n}$*. In this basis write*

$$E' = \begin{pmatrix} a & z^\dagger \\ z & D \end{pmatrix}, \qquad D = C + iK, \tag{3}$$

*where* $C$ *is a real symmetric* $2 \times 2$ *matrix and* $K$ *is real antisymmetric. Let* $c_1, c_2$ *be the eigenvalues of* $C$*, and put* $s = c_1 + c_2$*. Then*

$$\mathrm{Tr}(E) - t \geq z^\dagger D^+ z + \frac{(c_1 - c_2)^2}{s + t} \geq 0, \tag{4}$$

*where* $D^+$ *is the Moore–Penrose inverse. Positivity also requires* $z \in \operatorname{ran} D$*.*

*Proof.* In the adapted basis, the antisymmetric part has support only in the plane orthogonal to $\mathbf{n}$ and its two nonzero eigenvalues are $\pm t/2$ after multiplication by $i$. Hence $\det D = \det C - t^2/4$. Since $E' \geq 0$, the principal block $D$ is positive semidefinite and the generalized Schur-complement condition gives $z \in \operatorname{ran} D$ and $a \geq z^\dagger D^+ z$. Also $C \geq 0$ and $\det C \geq t^2/4$, so $s \geq t$. The identity $(c_1 - c_2)^2 = s^2 - 4 \det C$ therefore implies

$$(c_1 - c_2)^2 \leq s^2 - t^2 = (s - t)(s + t).$$

Finally, $\mathrm{Tr}(E) - t = a + s - t$. Combining these inequalities proves Eq. (4).

**Lemma 5.4** (Saturation for rank-one effects)**.** *If* $E$ *has rank one and* $\mathbf{b} \neq 0$*, the first inequality in Eq.* (4) *is an equality. In the adapted basis,* $a = 0$*,* $z = 0$*, and* $\det C = t^2/4$*.*

*Proof.* Write $E = \lambda |\psi\rangle\langle\psi|$ with $|\psi\rangle = \mathbf{x} + i\mathbf{y}$ and real vectors $\mathbf{x}, \mathbf{y}$. The antisymmetric part is proportional to $\mathbf{y}\mathbf{x}^T - \mathbf{x}\mathbf{y}^T$, so its axial direction $\mathbf{n}$ is perpendicular to both $\mathbf{x}$ and $\mathbf{y}$. Consequently $\langle \mathbf{n}|E|\mathbf{n}\rangle = 0$, and positivity of the rank-one matrix gives $E'|\mathbf{n}\rangle = 0$, hence $a = z = 0$. The lower block $D$ has rank one, so $\det D = 0$. Since $\det D = \det C - t^2/4$, we obtain $\det C = t^2/4$. The first inequality in Eq. (4) is therefore saturated. The nonnegative expression need not vanish for an anisotropic rank-one effect.

**Proposition 5.5** (Exact rank-one lift of five imaginarity vectors)**.** *Let* $\mathbf{b}_k \neq 0$ *for* $k = 1, \ldots, 5$*, put* $t_k = \|\mathbf{b}_k\|_2$ *and* $\mathbf{n}_k = \mathbf{b}_k / t_k$*, and assume* $\sum_k \mathbf{b}_k = 0$*. For each* $k$*, choose an oriented orthonormal frame* $F_k = (\mathbf{e}_{k1}, \mathbf{e}_{k2}) \in \mathbb{R}^{3\times 2}$ *with* $F_k^T F_k = I_2$*,* $F_k^T \mathbf{n}_k = 0$*, and* $\mathbf{e}_{k1} \times \mathbf{e}_{k2} = \mathbf{n}_k$*. Define*

$$B_k = -\frac{1}{2} \sum_{a,m,n=1}^{3} b_{ka}\, \epsilon_{amn} |m\rangle\langle n|.$$

*For arbitrary* $u_k \geq 0$ *and* $\theta_k \in [0, \pi)$*, let*

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$
$$C_k = \frac{t_k}{2} R(\theta_k) \begin{pmatrix} e^{u_k} & 0 \\ 0 & e^{-u_k} \end{pmatrix} R(\theta_k)^T.$$

*and set*

$$E_k = F_k C_k F_k^T + i B_k. \tag{5}$$

*Then each* $E_k$ *is positive and rank one, and its imaginarity vector is exactly* $\mathbf{b}_k$*. Conversely, every rank-one qutrit effect with nonzero imaginarity vector has this form. Consequently, the five effects form a POVM if and only if*

$$\sum_{k=1}^{5} F_k C_k F_k^T = I_3, \qquad \sum_{k=1}^{5} \mathbf{b}_k = 0. \tag{6}$$

*At a regular solution, where the six independent real components of Eq.* (6) *have Jacobian rank six with respect to the ten variables* $(u_1, \theta_1, \ldots, u_5, \theta_5)$*, the family of rank-one POVM lifts of the fixed vectors has local dimension four.*

*Proof.* In the oriented frame $(\mathbf{e}_{k1}, \mathbf{e}_{k2}, \mathbf{n}_k)$, the imaginary block $iB_k$ has transverse representation

$$\begin{pmatrix} 0 & -it_k/2 \\ it_k/2 & 0 \end{pmatrix}$$

and vanishes on the axial direction. The real part $F_k C_k F_k^T$ also vanishes on that direction. The transverse Hermitian block is $D_k = C_k + iK_k$, where $K_k$ is the displayed antisymmetric block. By construction, $C_k > 0$ and $\det C_k = t_k^2/4$, so $\det D_k = \det C_k - t_k^2/4 = 0$. Since $\operatorname{Tr} D_k = \operatorname{Tr} C_k > 0$, $D_k$ is positive semidefinite of rank one. This proves positivity, rank one, and the prescribed imaginary part.

Conversely, for a rank-one effect $E_k = |\psi_k\rangle\langle\psi_k|$ with $\mathbf{b}_k \neq 0$, write $\psi_k = \mathbf{x}_k + i\mathbf{y}_k$ with real vectors. The axial vector $\mathbf{b}_k$ is proportional to $\mathbf{x}_k \times \mathbf{y}_k$, so both real vectors lie in $\mathbf{n}_k^{\perp}$. Thus the real part has no axial row or column, and its transverse block $C_k$ is positive definite. The rank-one saturation lemma gives $\det C_k = t_k^2/4$. Every positive $2 \times 2$ matrix with this determinant has the stated eigenvalue parameterization for some $u_k \geq 0$ and $\theta_k$. Finally, POVM completeness separates into its real-symmetric and imaginary- antisymmetric parts, giving Eq. (6). The local dimension statement follows from the implicit function theorem at a rank-six regular point: ten internal variables minus six independent constraints leave four dimensions.

**Proposition 5.6** (An exact anisotropy budget for rank-one lifts)**.** *For a rank-one five-outcome POVM in the setting of Proposition 5.5, write*

$$T_k = R(\theta_k) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} R(\theta_k)^T,$$
$$\widetilde{T}_k = F_k T_k F_k^T, \qquad P_k = F_k F_k^T.$$

*Then POVM completeness is equivalent to the matrix equation*

$$\sum_{k=1}^{5} \frac{t_k}{2} \left( \cosh u_k \, P_k + \sinh u_k \, \widetilde{T}_k \right) = I_3. \tag{7}$$

*Taking the trace gives the exact identity*

$$\sum_{k=1}^{5} t_k \cosh u_k = 3, \qquad \sum_{k=1}^{5} t_k (\cosh u_k - 1) = \Delta, \tag{8}$$

*where* $\Delta = 3 - \sum_k t_k$*. In particular,*

$$\sum_{k=1}^{5} t_k u_k^2 \leq 2\Delta. \tag{9}$$

*For each outcome,*

$$|u_k| \leq \operatorname{arcosh}\left(1 + \frac{\Delta}{t_k}\right).$$

*If* $t_k \geq \tau > 0$ *for all* $k$*, then*

$$\max_k |u_k| \leq \sqrt{\frac{2\Delta}{\tau}}.$$

*Thus a small total defect forces every rank-one lift to be close to the isotropic transverse lift whenever the vector lengths are bounded away from zero.*

*Proof.* Since

$$R(\theta_k)\begin{pmatrix} e^{u_k} & 0 \\ 0 & e^{-u_k} \end{pmatrix} R(\theta_k)^T = \cosh u_k\, I_2 + \sinh u_k\, T_k,$$

the real part of Eq. (5) is exactly the summand in Eq. (7). The imaginary parts sum to zero precisely when $\sum_k \mathbf{b}_k = 0$, already assumed. Taking the trace and using $\operatorname{Tr} P_k = 2$ and $\operatorname{Tr} \widetilde{T}_k = 0$ yields $\sum_k t_k \cosh u_k = 3$. Subtracting $\sum_k t_k$ gives the second identity. Finally, $\cosh u - 1 \geq u^2/2$ proves the weighted bound. The individual bound follows because every summand in the defect budget is nonnegative. If $t_k \geq \tau$, then $t_k u_k^2 \leq 2\Delta$ gives the uniform estimate.

### 5.3 Zero-defect designs and kernel coordinates

**Theorem 5.7** (Exact classification of zero-defect rank-one five-outcome POVMs)**.** *Let $\{E_1, \ldots, E_5\}$ be a rank-one qutrit POVM with nonzero imaginarity vectors $\mathbf{b}_k = t_k \mathbf{n}_k$, and let $\Delta = 3 - \sum_k t_k$. Then $\Delta = 0$ if and only if*

$$\sum_{k=1}^{5} t_k \mathbf{n}_k = 0, \qquad \sum_{k=1}^{5} t_k \mathbf{n}_k \mathbf{n}_k^T = I_3.$$

*Equivalently, with $p_k = t_k/3$, the five weighted directions form a weighted spherical 2-design:*

$$\sum_k p_k = 1, \qquad \sum_k p_k \mathbf{n}_k = 0, \qquad \sum_k p_k \mathbf{n}_k \mathbf{n}_k^T = \frac{1}{3} I_3.$$

*Conversely, every such weighted design defines a rank-one qutrit POVM through the isotropic lift $u_k = 0$ in Eq. (5).*

*Proof.* If $\Delta = 0$, the exact anisotropy budget

$$\sum_k t_k (\cosh u_k - 1) = 0$$

and $t_k > 0$ imply $u_k = 0$ for every $k$. The real completeness condition therefore becomes

$$\frac{1}{2} \sum_k t_k (I_3 - \mathbf{n}_k \mathbf{n}_k^T) = I_3.$$

Since $\sum_k t_k = 3$, this is equivalent to $\sum_k t_k \mathbf{n}_k \mathbf{n}_k^T = I_3$. The imaginary completeness condition is $\sum_k t_k \mathbf{n}_k = 0$. Dividing by three gives the weighted design conditions.

Conversely, suppose the weighted design conditions hold and put $t_k = 3p_k$. Choose for each $\mathbf{n}_k$ an oriented orthonormal transverse frame $F_k$ and use the rank-one lift $C_k = (t_k/2) I_2$, i.e. $u_k = 0$. Its imaginary parts sum to zero by the first-moment condition, while

$$\sum_k F_k C_k F_k^T = \frac{1}{2} \sum_k t_k (I_3 - \mathbf{n}_k \mathbf{n}_k^T) = \frac{1}{2}(3 I_3 - I_3) = I_3.$$

Thus the lifted effects form a rank-one POVM.

**Corollary 5.8** (A one-vector parameterization of five-point weighted 2-designs)**.** *Let $p_k > 0$ and $\mathbf{n}_k \in S^2$ satisfy the weighted design conditions in Theorem 5.7. Put $V = (\sqrt{p_1}\mathbf{n}_1, \ldots, \sqrt{p_5}\mathbf{n}_5) \in \mathbb{R}^{3\times 5}$ and $q = (\sqrt{p_1}, \ldots, \sqrt{p_5})^T$. There is a unit vector $x \in \mathbb{R}^5$ such that*

$$V^T V = \frac{1}{3}(I_5 - qq^T - xx^T), \qquad q^T x = 0.$$

*In particular,*

$$p_k = \frac{1 - x_k^2}{4}, \qquad \sum_{k=1}^{5} x_k^2 = 1, \qquad \sum_{k=1}^{5} x_k \sqrt{1 - x_k^2} = 0. \tag{10}$$

*Conversely, any $x \in \mathbb{R}^5$ satisfying the last two conditions and $|x_k| < 1$ defines positive weights by $p_k = (1 - x_k^2)/4$ and admits a five-point weighted spherical 2-design, unique up to a common orthogonal transformation of $\mathbb{R}^3$.*

*Proof.* The design equations give $VV^T = I_3/3$ and $Vq = 0$. Hence $3V^TV$ is an orthogonal projection of rank three whose kernel is two-dimensional. One kernel vector is the unit vector $q$; choose a second unit kernel vector $x \perp q$. Thus $3V^TV = I_5 - qq^T - xx^T$. Comparing diagonal entries gives $3p_k = 1 - p_k - x_k^2$, hence the first relation in Eq. (10). Summing it gives $\sum_k x_k^2 = 1$, and $q^Tx = 0$ gives the final scalar equation.

Conversely, the two scalar conditions imply $\sum_k p_k = 1$ and $q^Tx = 0$. The matrix $P = I_5 - qq^T - xx^T$ is an orthogonal projection of rank three, and its diagonal entries are $3p_k$. Factor $P/3 = V^TV$ with $V \in \mathbb{R}^{3\times 5}$ and $VV^T = I_3/3$. Its columns have squared norms $p_k$; writing them as $\sqrt{p_k}\mathbf{n}_k$ gives unit vectors satisfying the weighted first- and second-moment conditions. Any two such factorizations differ by a common orthogonal transformation.

**Proposition 5.9** (Kernel-coordinate formula for the five-point design sensitivity)**.** *Let $x \in (-1, 1)^5$ satisfy*

$$\sum_{i=1}^{5} x_i^2 = 1, \qquad \sum_{i=1}^{5} x_i \sqrt{1 - x_i^2} = 0,$$

*and define $p_i = (1 - x_i^2)/4$. Let $G(x)$ be the $5 \times 5$ Gram matrix of the five physical generators $\mathbf{b}_i = 3p_i\mathbf{n}_i$ of the associated zero-defect rank-one POVM. Its entries are*

$$G_{ij}(x) = 3\sqrt{p_ip_j}\left(\delta_{ij} - \sqrt{p_ip_j} - x_ix_j\right). \tag{11}$$

*For every nonparallel pair $(i, j)$, put*

$$D_{ij} = G_{ii}G_{jj} - G_{ij}^2, \qquad T_{ijk} = \det G[\{i, j, k\}, \{i, j, k\}].$$

*Then its worst-case sensitivity is exactly*

$$\mathfrak{I}_0(x) = \min_{\substack{i<j \\ D_{ij}>0}} \frac{\sum_{k\neq i,j} \sqrt{T_{ijk}}}{2\sqrt{D_{ij}}}. \tag{12}$$

*Thus the zero-defect five-outcome problem is an explicit optimization over the two-constraint subset of $(-1, 1)^5$, with no reconstruction of the three-dimensional directions required.*

*Proof.* From the kernel representation,

$$V^TV = \frac{1}{3}(I_5 - qq^T - xx^T), \qquad q_i = \sqrt{p_i}.$$

Since $\mathbf{b}_i = 3p_i\mathbf{n}_i$ and the columns of $V$ are $\sqrt{p_i}\mathbf{n}_i$, the generator Gram matrix is $G_{ij} = 9p_ip_j\,\mathbf{n}_i \cdot \mathbf{n}_j = 9\sqrt{p_ip_j}(V^TV)_{ij}$, which gives Eq. (11). For a pair $(i, j)$, $D_{ij} = \|\mathbf{b}_i \times \mathbf{b}_j\|^2$. For a third index $k$, $T_{ijk} = \det G[\{i, j, k\}] = [\mathbf{b}_i \cdot (\mathbf{b}_j \times \mathbf{b}_k)]^2$. The zonotope facet normal is parallel to $\mathbf{b}_i \times \mathbf{b}_j$, and its support distance is

$$\frac{1}{2\|\mathbf{b}_i \times \mathbf{b}_j\|} \sum_{k\neq i,j} |\mathbf{b}_k \cdot (\mathbf{b}_i \times \mathbf{b}_j)|.$$

Substitution of the Gram determinants yields Eq. (12).

**Proposition 5.10** (Stability of the isotropic frame under small rank-one defect)**.** *Under the assumptions of Proposition 5.6, define the normalized-direction frame operator*

$$S = \sum_{k=1}^{5} t_k \mathbf{n}_k \mathbf{n}_k^T, \qquad \mathbf{n}_k = \mathbf{b}_k / t_k.$$

*Then*

$$\|S - (1-\Delta) I_3\|_\infty \le \Delta + \sqrt{6\Delta}. \tag{13}$$

*Equivalently, the isotropic transverse frame* $S_0 = \frac{1}{2}\sum_k t_k (I_3 - \mathbf{n}_k \mathbf{n}_k^T)$ *obeys*

$$\|I_3 - S_0\|_\infty \le \frac{1}{2}\left(\Delta + \sqrt{6\Delta}\right).$$

*In particular, along any sequence of rank-one five-outcome POVMs with* $\Delta \to 0$*, the weighted direction frame satisfies* $S \to I_3$ *in operator norm, without requiring a uniform lower bound on the individual* $t_k$*.*

*Proof.* Let $P_k = I_3 - \mathbf{n}_k \mathbf{n}_k^T$ and $\widetilde{T}_k = F_k T_k F_k^T$. The exact completeness equation in Proposition 5.6 gives

$$I_3 - S_0 = \frac{1}{2}\sum_k t_k \left[ (\cosh u_k - 1) P_k + \sinh u_k \, \widetilde{T}_k \right].$$

Since $\|P_k\|_\infty = \|\widetilde{T}_k\|_\infty = 1$,

$$\begin{aligned} 2\|I_3 - S_0\|_\infty &\le \sum_k t_k (\cosh u_k - 1) \\ &\quad + \sum_k t_k \sinh u_k. \end{aligned}$$

The first sum equals $\Delta$. By Cauchy–Schwarz,

$$\begin{aligned} \sum_k t_k \sinh u_k &= \sum_k \sqrt{t_k(\cosh u_k - 1)}\sqrt{t_k(\cosh u_k + 1)} \\ &\le \sqrt{\Delta \sum_k t_k (\cosh u_k + 1)} \\ &= \sqrt{\Delta(3 + \sum_k t_k)} \\ &= \sqrt{\Delta(6 - \Delta)} \le \sqrt{6\Delta}. \end{aligned}$$

Finally, $S_0 = \frac{1}{2}[(\sum_k t_k) I_3 - S] = \frac{1}{2}[(3-\Delta) I_3 - S]$, so $2(I_3 - S_0) = S - (1-\Delta) I_3$. This proves the result.

### 5.4 General effect defects and rank-two effects

**Proposition 5.11** (A frame-defect identity for qutrit POVMs)**.** *Let* $\{E_k\}_{k=1}^m$ *be a qutrit POVM and assume that every* $\mathbf{b}_k \neq 0$*. Write* $E_k = A_k + iB_k$*, set* $t_k = \|\mathbf{b}_k\|_2$*,* $\mathbf{n}_k = \mathbf{b}_k / t_k$*, and* $P_k = I - \mathbf{n}_k \mathbf{n}_k^T$*. In the real orthogonal basis adapted to* $\mathbf{n}_k$*, decompose*

$$A_k = a_k \mathbf{n}_k \mathbf{n}_k^T + \mathbf{n}_k \mathbf{z}_k^T + \mathbf{z}_k \mathbf{n}_k^T + \frac{s_k}{2} P_k + H_k,$$

*where* $\mathbf{z}_k \perp \mathbf{n}_k$*,* $s_k = \mathrm{Tr}(P_k A_k)$*, and* $H_k = P_k H_k P_k$ *is traceless on the transverse plane. Define*

$$\Delta = \sum_k \delta_k, \quad \delta_k = \mathrm{Tr}(E_k) - t_k, \quad S = \sum_k t_k \mathbf{n}_k \mathbf{n}_k^T.$$

*Then*

$$\begin{aligned} S - I = 3\sum_k a_k \mathbf{n}_k \mathbf{n}_k^T - \sum_k \delta_k \mathbf{n}_k \mathbf{n}_k^T \\ + 2\sum_k (\mathbf{n}_k \mathbf{z}_k^T + \mathbf{z}_k \mathbf{n}_k^T) + 2\sum_k H_k \\ - \Big(\sum_k a_k\Big) I. \end{aligned} \tag{14}$$

*Moreover, with* $s_k = t_k + \delta_k - a_k$,

$$\begin{aligned} \|S - I\|_\infty \le \sum_k \Big[ 4a_k + \delta_k + 2\sqrt{a_k s_k} \\ + \sqrt{(\delta_k - a_k)(s_k + t_k)} \Big]. \end{aligned} \tag{15}$$

*Proof.* In the adapted basis, write the real part in $1 + 2$ block form. Positivity gives $a_k \ge 0$, $\delta_k - a_k = s_k - t_k \ge 0$, and the generalized Schur complement gives $\|\mathbf{z}_k\|_2^2 \le a_k s_k$. If $c_{k,1}, c_{k,2}$ are the eigenvalues of the transverse block, then $\det C_k \ge t_k^2/4$, whence

$$|c_{k,1} - c_{k,2}|^2 \le (s_k - t_k)(s_k + t_k) = (\delta_k - a_k)(s_k + t_k).$$

Now sum the block decompositions and use $\sum_k A_k = I$ and $\sum_k \operatorname{Tr}(E_k) = 3$. The scalar part gives Eq. (14). For the norm bound, use $\|\sum a_k n_k n_k^T\|_\infty \le \sum a_k$, $\|\sum \delta_k n_k n_k^T\|_\infty \le \sum \delta_k$, $\|n_k z_k^T + z_k n_k^T\|_\infty = \|z_k\|_2$, and $2\|H_k\|_\infty = |c_{k,1} - c_{k,2}|$. The preceding inequalities yield Eq. (15).

**Lemma 5.12** (Schur-complement saturation for rank-two effects)**.** *Let* $E \ge 0$ *be a qutrit effect with rank two and nonzero imaginarity vector* $\mathbf{b}$*. In the basis adapted to* $\mathbf{n} = \mathbf{b}/\|\mathbf{b}\|_2$*, write*

$$E' = \begin{pmatrix} a & z^\dagger \\ z & D \end{pmatrix}, \qquad D = C + iK.$$

*If* $D$ *is positive definite, then*

$$a = z^\dagger D^{-1} z, \qquad \operatorname{Tr}(E) - t = z^\dagger D^{-1} z + (s - t), \tag{16}$$

*where* $t = \|\mathbf{b}\|_2$ *and* $s = \operatorname{Tr} C$*. In particular, the trace defect splits into an axial-coupling cost and a transverse-anisotropy cost.*

*Proof.* Since $D > 0$, the block determinant identity gives

$$\det E' = \det D\,(a - z^\dagger D^{-1} z).$$

The rank-two condition implies $\det E' = 0$, and hence $a = z^\dagger D^{-1} z$. Also $\operatorname{Tr}(E) = a + s$ and positivity of $D$ gives $s \ge t$ from $\det D = \det C - t^2/4 \ge 0$. Substitution proves Eq. (16).

**Proposition 5.13** (Defect–frame stability for qutrit POVMs)**.** *Let* $\{E_k\}_{k=1}^m$ *be a qutrit POVM with every* $\mathbf{b}_k \ne 0$*, let* $\mathbf{b}_k$ *be its imaginarity vectors, and set* $t_k = \|\mathbf{b}_k\|_2$*,* $\delta_k = \operatorname{Tr}(E_k) - t_k$*, and* $\Delta = \sum_k \delta_k = 3 - \sum_k t_k$*. For nonzero* $\mathbf{b}_k$*, put* $\mathbf{n}_k = \mathbf{b}_k/t_k$ *and* $S = \sum_k t_k \mathbf{n}_k \mathbf{n}_k^T$*. In the adapted block decomposition of Proposition 5.11, let* $A = \sum_k a_k$*. Then*

$$\begin{aligned} 0 \le A \le \Delta, \qquad \|S - I\|_\infty \le 4A + \Delta + 2\sqrt{A(3 - A)} \\ + \sqrt{(\Delta - A)(6 - \Delta - A)}. \end{aligned} \tag{17}$$

*In particular,* $\Delta = 0$ *implies* $S = I$*.*

*Proof.* For each effect, positivity gives $0 \le a_k \le \delta_k$ and $s_k = t_k + \delta_k - a_k$. Hence $\sum_k s_k = 3 - A$ and $\sum_k (s_k + t_k) = 6 - \Delta - A$. Apply the frame-defect bound (15) and Cauchy–Schwarz separately:

$$\sum_k \sqrt{a_k s_k} \le \sqrt{\Big(\sum_k a_k\Big)\Big(\sum_k s_k\Big)} = \sqrt{A(3-A)},$$

$$\sum_k \sqrt{(\delta_k - a_k)(s_k + t_k)} \le \sqrt{(\Delta - A)(6 - \Delta - A)}.$$

This proves Eq. (17). If $\Delta = 0$, then $A = 0$ and the right-hand side vanishes, so $S = I$.

**Corollary 5.14** (A defect lower bound allowing zero imaginarity vectors)**.** *Let $\{E_k\}_{k=1}^m$ be an arbitrary qutrit POVM, allowing $\mathbf{b}_k = 0$ for some outcomes. Define*

$$S = \sum_{\mathbf{b}_k \neq 0} \frac{\mathbf{b}_k \mathbf{b}_k^T}{\|\mathbf{b}_k\|_2},$$
$$\gamma = \|S - I\|_\infty, \qquad \Delta = 3 - \sum_k \|\mathbf{b}_k\|_2.$$

*Then*

$$\Delta \ge \left[\frac{\sqrt{18 + 20\gamma} - 3\sqrt{2}}{10}\right]^2. \tag{18}$$

*Proof.* Split the outcomes into $\mathcal{N} = \{k : \mathbf{b}_k \neq 0\}$ and $\mathcal{Z} = \{k : \mathbf{b}_k = 0\}$. For $k \in \mathcal{Z}$, the effect is real and positive; write $E_k = A_k^{(0)}$ and set $\Delta_0 = \sum_{k\in\mathcal{Z}} \operatorname{Tr} A_k^{(0)}$. For the nonzero-vector outcomes, use the adapted decomposition of Proposition 5.11. Let $\Delta_\mathcal{N} = \sum_{k\in\mathcal{N}} \delta_k$ and $A = \sum_{k\in\mathcal{N}} a_k$. Then $\Delta = \Delta_\mathcal{N} + \Delta_0$ and $0 \le A \le \Delta_\mathcal{N}$. Since $\sum_{k\in\mathcal{N}} A_k = I - \sum_{k\in\mathcal{Z}} A_k^{(0)}$, the exact frame-defect identity for the nonzero-vector outcomes is

$$\begin{aligned} S - I = {} & 3\sum_{k\in\mathcal{N}} a_k n_k n_k^T - \sum_{k\in\mathcal{N}} \delta_k n_k n_k^T \\ & + 2\sum_{k\in\mathcal{N}} (n_k z_k^T + z_k n_k^T) + 2\sum_{k\in\mathcal{N}} H_k - AI \\ & + 2R - \Delta_0 I, \qquad R = \sum_{k\in\mathcal{Z}} A_k^{(0)}. \end{aligned}$$

Because $R \ge 0$ and $\operatorname{Tr} R = \Delta_0$, its eigenvalues lie in $[0, \Delta_0]$, so $\|2R - \Delta_0 I\|_\infty \le \Delta_0$. Repeating the proof of Proposition 5.13 therefore gives

$$\begin{aligned} \gamma \le {} & \Delta_0 + 4A + \Delta_\mathcal{N} + 2\sqrt{A(3 - \Delta_0 - A)} \\ & + \sqrt{(\Delta_\mathcal{N} - A)(6 - \Delta - \Delta_0 - A)}. \end{aligned}$$

Using $A \le \Delta_\mathcal{N}$ and $\Delta_0 + \Delta_\mathcal{N} = \Delta$, the right-hand side is bounded by $5\Delta + 3\sqrt{2\Delta}$, by the same Cauchy–Schwarz estimate as above. Solving for $\Delta$ proves the claim.

**Corollary 5.15** (A rank-one defect lower bound, including zero vectors)**.** *If every effect is rank one, zero-imaginarity outcomes are allowed and*

$$\|S - I\|_\infty \le \Delta + \sqrt{\Delta(6 - \Delta)}. \tag{19}$$

*Consequently, if $\gamma = \|S - I\|_\infty$, then*

$$\Delta \ge \frac{3 + \gamma - \sqrt{9 + 6\gamma - \gamma^2}}{2}. \tag{20}$$

*Proof.* For every nonzero-vector rank-one effect, $a_k = 0$. Let $\Delta_0$ and $\Delta_{\mathcal{N}}$ denote the total defects of the zero- and nonzero-vector outcomes, respectively. The corrected zero-vector decomposition and the frame-defect bound give

$$\gamma \le \Delta_0 + \Delta_{\mathcal{N}} + \sqrt{\Delta_{\mathcal{N}}(6 - \Delta - \Delta_0)} \le \Delta + \sqrt{\Delta(6 - \Delta)}.$$

Solving the resulting quadratic inequality gives the stated lower bound.

**Corollary 5.16** (A global defect budget for qutrit POVMs)**.** *For a qutrit POVM $\{E_k\}_{k=1}^m$, write $E_k = A_k + iB_k$ and let $\mathbf{b}_k$ be its imaginarity vectors. For every nonzero $\mathbf{b}_k$, put $\mathbf{n}_k = \mathbf{b}_k/\|\mathbf{b}_k\|_2$. Then*

$$3 - \sum_{k=1}^{m} \|\mathbf{b}_k\|_2 \ge \sum_{\mathbf{b}_k \neq 0} \mathbf{n}_k^T A_k \mathbf{n}_k \ge 0. \tag{21}$$

*Proof.* Sum Eq. (2) over all nonzero vectors and use $\sum_k \mathrm{Tr}(E_k) = 3$ and $\sum_k A_k = I_3$. Terms with $\mathbf{b}_k = 0$ contribute $\mathrm{Tr}(A_k) \ge 0$ to the left-hand side and may be discarded.

**Proposition 5.17** (A volume-to-surface upper bound for the minimax sensitivity)**.** *Let $\mathbf{b}_1, \dots, \mathbf{b}_m$ span $\mathbb{R}^3$, and let*

$$Z = \sum_{k=1}^{m} [-\mathbf{b}_k/2, \mathbf{b}_k/2]$$

*be their centered zonotope. Define*

$$V_3 = \sum_{i<j<k} |\det(\mathbf{b}_i, \mathbf{b}_j, \mathbf{b}_k)|, \qquad A_2 = 2\sum_{i<j} \|\mathbf{b}_i \times \mathbf{b}_j\|_2.$$

*Then*

$$\underline{\mathfrak{I}}(\mathcal{E}) \le \frac{3V_3}{A_2} = \frac{3}{2} \frac{\sum_{i<j<k} |\det(\mathbf{b}_i, \mathbf{b}_j, \mathbf{b}_k)|}{\sum_{i<j} \|\mathbf{b}_i \times \mathbf{b}_j\|_2}. \tag{22}$$

*Equality holds if and only if every facet of $Z$ has the same distance from the origin.*

*Proof.* The volume and surface area of the zonotope are

$$\mathrm{Vol}(Z) = V_3, \qquad \mathrm{Area}(\partial Z) = A_2.$$

Decompose $Z$ into pyramids with common apex at the origin and bases given by its facets. If $h_f$ and $A_f$ are the distance and area of facet $f$, respectively, then

$$V_3 = \frac{1}{3} \sum_f h_f A_f, \qquad A_2 = \sum_f A_f.$$

The minimax sensitivity is the inradius $\min_f h_f$. It cannot exceed the area-weighted mean $\sum_f h_f A_f / A_2 = 3V_3/A_2$. Equality holds precisely when all positive-area facets have the same distance from the origin.

**Proposition 5.18** (Parameterization of the full-support rank-two five-outcome stratum)**.** *Consider a five-outcome qutrit POVM with $\mathrm{rank}\, E_1 = \cdots = \mathrm{rank}\, E_4 = 1$ and $\mathrm{rank}\, E_5 = 2$. Write $E_5 = R$ and $E_k = q_k^\dagger q_k$ for $k = 1, \dots, 4$, and stack the four row vectors into a matrix $Q \in \mathbb{C}^{4\times 3}$. Then*

$$Q^\dagger Q = I_3 - R.$$

*If $I_3 - R$ is positive definite, every such POVM has the representation*

$$Q = V(I_3 - R)^{1/2}, \qquad V^\dagger V = I_3, \tag{23}$$

*where $R$ is a rank-two effect satisfying $0 \le R < I_3$ and $V$ belongs to the complex Stiefel manifold $\mathrm{St}(3, 4)$. Conversely, every such pair $(R, V)$ defines a five-outcome POVM with the stated rank pattern, provided no row of $Q$ vanishes.*

*Proof.* Completeness gives $\sum_{k=1}^{4} q_k^\dagger q_k + R = I_3$, which is precisely $Q^\dagger Q = I_3 - R$. If $I_3 - R > 0$, the polar decomposition of $Q$ has the form $Q = V(Q^\dagger Q)^{1/2}$ with $V^\dagger V = I_3$, yielding Eq. (23). The rank-two positive-effect manifold has real dimension 8, while the complex Stiefel manifold of $4 \times 3$ isometries has real dimension 15. Quotienting the four independent row phases, which do not change the effects, gives a generic rank-two stratum of real dimension 19. By comparison, the generic all-rank-one five-outcome stratum has dimension 16. Conversely, the stated factorization implies $Q^\dagger Q + R = I_3$, so the five effects are positive and sum to the identity. The rank assumptions follow from $\operatorname{rank} R = 2$ and the nonvanishing of the four rows.

**Proposition 5.19** (The rank-two stratum accumulates on every rank-one five-outcome POVM)**.** *Let $\mathcal{E} = \{q_k^\dagger q_k\}_{k=1}^5$ be a rank-one qutrit POVM with all five effects nonzero. Then there is a sequence of five-outcome POVMs $\mathcal{E}^{(n)}$ of rank pattern $(1,1,1,1,2)$ converging to $\mathcal{E}$. Moreover, the guaranteed sensitivity is continuous under this convergence. Consequently,*

$$\sup_{\mathcal{F}:\,\operatorname{rank} F_1=\cdots=\operatorname{rank} F_4=1,\,\operatorname{rank} F_5=2} \underline{\mathfrak{I}}(\mathcal{F}) \geq \sup_{\mathcal{E}:\,\operatorname{rank} E_k=1} \underline{\mathfrak{I}}(\mathcal{E}). \tag{24}$$

*Proof.* Stack the five row vectors into $Q_0 \in \mathbb{C}^{5\times 3}$, so $Q_0^\dagger Q_0 = I_3$. Define

$$W_0 = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5/\sqrt{2} \\ q_5/\sqrt{2} \end{pmatrix} \in \mathbb{C}^{6\times 3}.$$

Then $W_0^\dagger W_0 = I_3$. Choose a generic matrix $X \in \mathbb{C}^{6\times 3}$ and, for sufficiently small $\epsilon > 0$, set

$$W_\epsilon = (W_0 + \epsilon X)\big[(W_0 + \epsilon X)^\dagger (W_0 + \epsilon X)\big]^{-1/2}.$$

Thus $W_\epsilon^\dagger W_\epsilon = I_3$ and $W_\epsilon \to W_0$. For generic $X$, the last two rows of $W_\epsilon$ are linearly independent, while the first four rows are nonzero. Writing $W_\epsilon = (Q_\epsilon^T, Z_\epsilon^T)^T$, the effects $q_{k,\epsilon}^\dagger q_{k,\epsilon}$ $(k = 1,\dots,4)$ and $Z_\epsilon^\dagger Z_\epsilon$ therefore have ranks $(1,1,1,1,2)$ and sum to $I_3$. They converge to the original five effects.

For continuity, let $\mathbf{b}_k$ and $\mathbf{c}_k$ be the three-component spin-vector representations of two five-outcome POVMs. Their directional responses satisfy

$$\sup_{\|\mathbf{a}\|=1} |h_{\mathcal{E}}(\mathbf{a}) - h_{\mathcal{F}}(\mathbf{a})| \leq \frac{1}{2} \sum_{k=1}^{5} \|\mathbf{b}_k - \mathbf{c}_k\|_2.$$

Taking minima over the unit sphere shows that the guaranteed sensitivities differ by no more than the same quantity. This proves continuity and Eq. (24).

**Proposition 5.20** (Rank-two effects are coarse-grainings of two rank-one outcomes)**.** *If a five-outcome qutrit POVM has rank pattern $(1,1,1,1,2)$, the spectral decomposition gives a six-outcome rank-one refinement*

$$\mathcal{E}' = \{E_1, E_2, E_3, E_4, \lambda_1|u_1\rangle\langle u_1|, \lambda_2|u_2\rangle\langle u_2|\},$$

*where $E_5 = \lambda_1|u_1\rangle\langle u_1| + \lambda_2|u_2\rangle\langle u_2|$. For every input state, the total-variation distance of the original five-outcome distributions does not exceed that of the refined six-outcome distributions. Hence*

$$\underline{\mathfrak{I}}(\mathcal{E}) \leq \underline{\mathfrak{I}}(\mathcal{E}'). \tag{25}$$

*Proof.* The original fifth outcome is obtained by merging the last two refined outcomes. Total variation is contractive under this deterministic classical postprocessing, for both $\rho$ and $\rho^*$. The pointwise inequality survives division by $\mathcal{M}(\rho)$ and taking the infimum over states.

## 6 What the numerical certificate establishes

The interval program `certify_five_outcome_v53.py` encloses the exact trigonometric construction using real interval operations on the real and imaginary components. Its inputs contain no binary floating-point approximations. The final decimal endpoints are proposed from the interval endpoints and then accepted only if outward interval comparisons verify the claimed enclosure. This certifies the stated five-outcome improvement within the interval implementation, not the global five-outcome optimum or every analytic theorem.

The additional `verify_baselines_v53.py` checks the beta-integral Haar formula numerically and compares facet responses with direct measurement probabilities. Monte Carlo agreement checks normalization and implementation, rather than proving a worst-case bound. The external zonoid approximation and empirical-process inputs remain cited mathematical results; these programs do not independently prove them or establish literature priority.

## 7 Reproduction

The file `reproduce_paper_examples_v53.py` requires Python 3, NumPy, SciPy, and mpmath. Run

```
python reproduce_paper_examples_v53.py
```

The program prints the dimension-dependent minimax constants, the continuous-frame Monte Carlo checks, the polyhedral qutrit sensitivities, the ten facet values of the five-outcome construction, and the residuals of the isometry and closure constraints. The five-outcome construction is evaluated with 60-digit arithmetic; the Monte Carlo and polyhedral checks use double precision. No numerical search is used to establish the analytic theorems in the main article.